



Optimized machine learning enabled intrusion detection 2 system for internet of medical things

Zhenyang Sun, Gangyi An, Yixuan Yang, Yasong Liu^{*}

King's college London, Strand, London, WC2R 2LS, United Kingdom

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ABSTRACT

This research paper proposes an intrusion detection system that combines particle swarm optimization and AdaBoost algorithms to classify and detect malware-related records in innovative health app platforms. The research employs the NSL KDD dataset, consisting of 125,973 instances and 41 features, with data split into training (20 %) and testing (80 %) sets. A feature selection process using Particle Swarm Optimization identifies 12 relevant features for intrusion detection. The intrusion detection system effectively categorizes various attack types, including Denial of Service (DoS), User-to-root (U2R), Root-to-local (R2L), and Probe attacks. Regarding classifier performance, AdaBoost exhibits the highest recall value (0.966667), highlighting its strong intrusion detection capabilities. The experimental results demonstrate that the PSO-AdaBoost approach achieves superior accuracy, precision, and recall in intrusion detection. By integrating machine learning-based intrusion detection systems into smart health apps, healthcare providers can enhance patient care, reduce costs, and ensure the confidentiality of sensitive medical information. This research highlights the potential of ML-IDSs in strengthening the security infrastructure surrounding medical IoT devices and improving patient outcomes in the interconnected world of the Internet of Medical Things.

1. Introduction

The rapid advancement of the Internet of Things (IoT) has led to the emergence of the Internet of Medical Things (IoMT), revolutionizing healthcare by connecting medical devices and systems to the Internet [1, 2]. IoMT offers numerous benefits, including remote patient monitoring, real-time health data collection, and improved healthcare delivery. However, the increasing connectivity also brings inherent security risks, making protecting sensitive medical data and ensuring the safety of patients a paramount concern [3]. Intrusion Detection Systems (IDS) are crucial in safeguarding IoMT environments by monitoring network traffic, identifying potential security threats, and providing early warning of malicious activities [4]. Traditional IDS solutions, though effective in general settings, may not be optimized to handle the unique challenges posed by the IoMT ecosystem. This is where machine learning techniques can be leveraged to develop optimized IDS solutions capable of addressing the specific requirements of the IoMT environment. Machine learning has shown remarkable success in various domains, including image recognition, natural language processing, and anomaly detection [5]. By harnessing the power of machine learning

algorithms, an optimized IDS for IoMT can learn and adapt to evolving threats, identify patterns, and detect anomalies in real-time [6]. This enables healthcare providers to proactively respond to security breaches, protect patient privacy, and ensure the uninterrupted functioning of critical medical devices.

The Internet of Medical Things (IoMT) is rapidly expanding, with an estimated 30.09 billion devices in use by 2025 and a projected increase to 100.24 billion by 2030 [7]. IoMT offers immediate medical treatment, accurate diagnoses, and continuous real-time patient monitoring, particularly in hospital settings [8]. However, the increasing connectivity of medical devices poses security risks, and malicious attacks can expose vital patient medical information, jeopardizing lives. Optimized machine learning-enabled Intrusion Detection Systems (IDS) are being developed specifically for the IoMT environment to address these concerns. These systems leverage machine learning algorithms to detect and prevent real-time security breaches. By analyzing network traffic and identifying anomalies, they safeguard patient privacy and ensure the uninterrupted functioning of critical medical devices. The healthcare industry recognizes the potential of IoMT to enhance patient care and reduce costs [9,10]. However, privacy protection and security measures

^{*} Corresponding author.

E-mail address: yasongliu071@gmail.com (Y. Liu).

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are paramount due to the sensitive nature of medical data. Neglecting security during IoMT development can result in significant financial consequences. Therefore, it is crucial to implement robust security measures from the outset to mitigate risks. By advancing the field of IoMT security through optimized IDS solutions, incorporating machine learning algorithms, and addressing privacy concerns, trust in connected medical devices can be established. This will promote widespread adoption of IoMT and maximize its benefits in healthcare.

Hackers constantly target confidential medical information and exploit Internet of Things (IoT) devices to create botnets [11]. Botnets are networks of compromised computers used to carry out various malicious activities. Some botnets are particularly adept at launching distributed denial of service (DDoS) attacks, which overload online services with excessive traffic from multiple sources. Unsecured IoT devices can be easily added to these botnets, allowing cybercriminals to launch attacks on other websites. Such intrusions compromise the targeted systems' availability, security, and privacy. When malicious users gain unauthorised access to a computer system, its defences are immediately compromised. Security breaches can take the form of network intrusions or host intrusions. Network intrusions involve allowing malicious packets into a network to initiate DoS attacks or break into machines. Denial-of-service attacks aim to prevent authorized users from accessing their computers [12]. Symptoms of an intrusion include unusual system behavior, unexpected crashes, changes to the kernel's data structures, and degraded network performance. Intrusion prevention measures safeguard computer systems, such as user authentication, programming error prevention, and data encryption. However, an additional layer of defense in the form of intrusion detection is necessary. Intrusion detection refers to monitoring systems or networks to identify any unauthorised access, suspicious activities, or data tampering. Effective intrusion detection relies on models that classify normal system behavior, techniques that compare actual activities to these models to identify abnormal or intrusive behavior, and available resources [13]. By implementing intrusion detection, computer systems can enhance their protection against potential cyber threats. Machine learning techniques for intrusion detection can be classified into three categories: supervised, unsupervised, and semi-supervised learning [14].

In unsupervised learning, input samples lack pre-labelled classes, allowing the model to learn patterns independently. Supervised learning requires a labelled dataset to train the model accurately. However, marking extensive network and host data can be costly. Semi-supervised learning methods leverage labelled and unlabeled instances, effectively using abundant unlabeled data with minimal labelled data. Trained classifiers or models categorize test patterns based on the prioritized features during training. Machine learning-based intrusion detection systems, known as ML-IDSs, automate identifying attacks, reducing the need for human intervention and making them increasingly valuable in computer security [15]. Building a classification model using machine learning techniques necessitates a dataset, and it is advisable to construct and test classifiers on various datasets. Well-known and frequently used datasets for intrusion detection include the KDD99 database, DARPA 1998/1999 database, and ISCX 2012 IDS dataset [16].

This research aims to develop an advanced intrusion detection system (IDS) by leveraging the Particle Swarm Optimization (PSO) and AdaBoost algorithms. The primary objective is to optimize the feature selection process and enhance the performance of intrusion detection and classification in computer networks. In this research, we investigate and implement the PSO algorithm to efficiently select relevant features from the input dataset, focusing on those crucial for accurate intrusion detection. Multiple classification models, including AdaBoost, K-Nearest Neighbors (KNN), and Naïve Bayes, which we trained and tested to detect and classify malware-related records. In experimental work, we utilize the NSL KDD dataset, a widely recognized benchmark, for evaluating the proposed system's effectiveness. The ultimate objective is to implement the PSO-AdaBoost IDS, achieving the highest accuracy, recall

accuracy, precision, and recall levels for intrusion detection and classification tasks. In this research, we also analyze the results and provide valuable insights into the effectiveness of the PSO-AdaBoost approach, paving the way for developing more robust intrusion detection systems in various domains while enhancing network security and response capabilities.

2. Literature survey

2.1. Review of optimization algorithms for IDS

The cloud-based intrusion detection model that Sun et al. developed uses an improved version of the backpropagation (BP) algorithm [17]. This model is created by combining the BP algorithm's powers of doing local searches with the PSO algorithm's capabilities of performing global optimizations. We present a PSO method to optimize the value of the initial weight and threshold of BP within the context of the momentum and adaptive learning rate approach. This helps to ensure that the BP does not fall into a local optimum and that its speed of network convergence is increased to the greatest possible extent. The results of the studies demonstrate that the recommended model has a better average detection rate, and it is possible to use it to identify intrusions in the cloud. The comparison of optimization algorithms for Intrusion Detection Systems (IDS) along with their advantages and disadvantages, as mentioned in Table 1

Seth et al. have developed a method for the network intrusion detection system that uses key feature selection based on binary Grey Wolf Optimization (GWO) and a neural network classifier [18]. The installation of the IDS may be carried out at any critical node in the network if desired. It is possible to utilize GWO to remove unnecessary features from the dataset, which in turn decreases the total size of the dataset, reducing the amount of time needed to train the classifier and the amount of space required to keep the dataset. When doing simulations using the NSL-KDD dataset, we discovered that utilizing a more condensed feature set improves the accuracy of the intrusion detection approach.

In 2016, Jaiswal et al. [19] developed an enhanced method of intrusion detection by using a data mining approach. The intrusion detection system monitors the user's activities. The system watches the user's actions, notifying the administrator of potentially harmful behavior. In this experiment, we train a KNN classifier to find invaders by use the KDD Cup dataset to categorize the various forms of invasion. We prepare a KNN classifier to find invaders using the ACO method.

The correctness of simulated data and a performance statistic is known as the False Alarm Rate (FAR). The outputs of this methodology are more exact when compared to the way that is being used at the moment.

Principal component analysis was used by Ahmad and e Amin [20] to limit the characteristics that should be used. The features were first projected into an eigenspace, and after that, the ones whose deviations were found to be the greatest were selected for further study. The features with the highest eigenvalue may provide the classifier with the best sensitivity, but those characteristics will still be employed (i.e., eigenvalues). It is not sufficient to choose the features that have the highest eigenvalues; instead, an optimization technique is required to discover which parts, after being transformed, retain the most significant number of their original properties. This is because an element of optimization is involved in selecting the collection of converted characteristics that most effectively differentiate the object under consideration. Studies have been conducted using the evolutionary optimization approach known as the genetic algorithm (GA), to determine which characteristics may be optimized for more significant differentiation. Another form of optimization, particle swarm optimization or PSO for short, draws its motivation from research on the behavior of animals and birds. PSO may be beneficial in some circumstances more significantly than GA in some instances.

Table 1

Comparison of optimization algorithms for intrusion detection systems (IDS).

Optimization Algorithm	Advantages	Disadvantages	Purpose of Proposed Model
Genetic Algorithms	- Global search capability. - Can handle non-linear optimization problems. - Adaptive and capable of evolving solutions. - Effective for feature selection.	- Can be computationally expensive. - May converge to suboptimal solutions. - Requires tuning of parameters. - Randomness may affect performance.	To enhance the feature selection process and improve the efficiency of the IDS by exploring a broader search space.
Particle Swarm Optimization (PSO)	- Simple implementation. - Capable of global optimization. - Effective for parameter tuning. - Converges quickly to optimal solutions in some cases.	- May get stuck in local optima. - Performance depends on the choice of parameters and initialization. - Limited exploration of the search space.	To optimize the parameters of machine learning models used in the IDS, improving its detection accuracy.
Ant Colony Optimization (ACO)	- Inspired by nature and adaptable to dynamic environments. - Effective for routing and path optimization. - Scalable and parallelizable. - Can handle discrete problems.	- May require fine-tuning of parameters. - May suffer from slow convergence. - Performance depends on the construction of pheromone matrices.	To develop an IDS capable of adapting to evolving attack patterns in dynamic networks.
Simulated Annealing (SA)	- Good at escaping local optima. - Allows for exploring suboptimal solutions initially. - Can be used for continuous and discrete optimization problems.	- Requires careful tuning of the cooling schedule. - May be slow to converge compared to other methods. - Vulnerable to getting stuck in local minima.	To improve the robustness of the IDS against sophisticated evasion techniques and optimize its parameter settings.
Tabu Search	- Effective at escaping local optima. - Can handle discrete and continuous optimization. - Adaptable and capable of intensification and diversification.	- Requires careful tuning of tabu list parameters. - High sensitivity to initial solutions. - Computationally expensive for large search spaces.	To develop an IDS that can dynamically adapt to changes in network traffic and intrusion patterns.

Tabatabaefar et al. [21] suggest creating two sets of antibodies, one positive and one negative, to identify standard and assault samples in their proposal for an AIS-based intrusion detection system. These antibodies would be used to differentiate between the two types of models. The ideas of positive and negative selection are used in this system to facilitate the production of primary detectors. Consequently, training the more modern sensors with the more conventional PSO can result in a higher detection rate. Antibody radii are likewise constantly established by the algorithms responsible for their development and training. The simulation results indicate that the recently created approach obtains a true positive rate of 99.1 % while only producing a false positive rate of 1.9 % of the time.

Using the periodicity, unpredictability, and sensitivity of the chaos operator to the initial conditions, The author's Yang et al. [22], create an improved version of the particle swarm optimization IPSO technique. Particle swarm optimization (PSO) is used to optimize the value of the RBF kernel function parameter g and the penalty factor C . This helps to increase convergence and accuracy. Experiments have demonstrated that the IPSO-SVM intrusion detection model is superior to the PSO-SVM and the GA-SVM models in identifying intrusions. This argument is driven home by the higher performance of IPSO-superiority SVMs over both models.

Hosseini et al. [23], discuss the significance of a DoS attack intrusion detection system based on support vector regression and how it can be enhanced by combining two additional methods, such as an ant colony and a firefly. They explain how this can be accomplished. Hosseini and colleagues discuss, among other things, the significance of using an intrusion detection system for DoS attacks that are based on support vector regression (2015). The firefly technique utilizes a local search to move the ants, and its overall effectiveness may be enhanced by removing random variables from the equation. Using the KDD Cup 99 evaluation framework allowed for the intrusion detection system to be put through its paces and given a thorough assessment. Only 9 of the original forty-one available homes have been selected as potential rentals by our team. The recommended approach achieves an accuracy of 99.57 percent and a detection error rate of only 0.0064 percent.

Kulshrestha et al. [24], provide an innovative strategy to find which

feature subset and classifier function most effectively with a particular dataset. If the issue were sufficiently large, searching for a solution in a large area would be necessary. This would be the case because of the previous sentence. Therefore, a powerful meta-heuristic known as the cuckoo search CS approach has been used in the search to select a feature subset and classifier that minimizes both the dimensionality of the feature vector and the rate of classification error. This was done to establish a feature subset and classifier that best fits the criteria. The CSFCS has been validated and shown to be accurate by employing various datasets considered the standards in their respective industries. CSFCS is a flexible and realizable method for producing CS, which is necessary for the work because it improves classification accuracy with a smaller collection of features. To create CS, CSFCS was shown to be an option. This is because CSFCS does not need a complete suite of functions and may be readily modified to suit individual needs.

2.2. Classification methods for IDS

Zhao and Zhang [25], have developed a novel intrusion detection technique based on augmented K-means to respond to the specific features of cloud computing and based on augmented K-means to react to the particular characteristics of cloud computing and the unique security needs of cloud computing. A clustering algorithm and a technique for distributed intrusion detection may be developed using this technology. It can identify both predicted and unanticipated assaults in cloud computing environments. The simulation findings indicate that using this strategy might hasten the process of intrusion detection while simultaneously reducing the rates of false positives and false negatives.

Chen et al. [26], introduced an approach that combines rough sets with data mining to improve the detection efficacy of intrusion detection while simultaneously reducing the number of false alarms. This method was compared to established forms of intrusion detection. The first phase is organizing the data gathered, followed by normalizing the values of the collected variables and then processing the nominal variables discretely like that in which the attributes of the property up and down approximation set may be employed in the Pawlak attribute weight rough set technique to conduct an attribute reduction on the final

result set, the last result set will be described here. The generation of high-confidence association rules, which are then imported into the power set once attribute reduction has been completed, is made possible thanks to this process. Experiments show that the detection technique that uses both rough sets and data mining results in an improvement of around 20 % in detection efficiency. The rate at which invasions are uncovered accelerates almost linearly in proportion to the overall number of attacks.

Jing and Bin [27], have developed and implemented similar approaches for detecting intrusions into a network. They do this using a limited Boltzmann machine trained using a relevant deep-learning model. In this study, we analyzed past research in the area to determine whether or not it would be feasible to use deep learning to network IDS. The technology used to identify intrusions into networks is now undergoing development with the assistance of relevance depth learning to accomplish greater detection accuracy. The findings of the simulation show that the approach to network intrusion detection that relies on relevance deep learning performs better, both in terms of the average detection rate and the average false detection rate for unknown intrusions and assaults. Specifically, the average detection rate for unknown intrusions and assaults is lower. The results of the experiments support the claims made about the resilience and effectiveness of the recently proposed approach.

Yuan et al. [28], accomplished adaptive network intrusion detection by applying the Two Layers Multi-class Detection (TLMD) method. Utilizing this method in conjunction with the C5.0 method and the Naive Bayes algorithm results in an improvement not only to the detection rate but also to the false alarm rate. The TLMD method addresses all of the challenges specific to data mining, each of which can be considered unique. Handling contiguous properties is one of these issues, and it also reduces the amount of noise in the training dataset. The process can handle this complicated situation. We compare the recently developed TLMD strategy to the methods that are considered to be state-of-the-art by using the detection rate, accuracy, and false alarm rate on the KDD Cup99 benchmark dataset for intrusion detection. Even when applied to an unbalanced dataset, the one-of-a-kind TLMD method appears to have a low false alarm rate and a high detection rate. This is the case even though the dataset is not balanced. Every piece of evidence supports this judgment.

A risk-based warning prioritizing technique was employed by Chakir et al. [28], for the research that they conducted. The model can assess the warning's severity, considering various decision factors such as priority, reliability, and asset value indicators to determine the appropriate intensity of the warning notification. This improves Snort's ability to detect intrusions and presents the security administrator with only those alerts that pose a genuine risk, reducing the number of false positives and time spent analyzing them. The goal is to determine the significance of individual IDS alerts in terms of the overall security of an information system. This determines the importance of personal IDS alerts in terms of the general safety of an information system. We use the pattern-matching method and the KDD Cup 99 Dataset to evaluate the model.

Gaikwad and Thool [29], have developed a brand-new intrusion detection strategy using ensemble machine learning. It is helpful to employ the Bagging approach of ensembles while setting IDS. This technique utilizes the REP Tree as the foundation class. The classification accuracy may be improved while simultaneously lowering the number of false positives if the proper attributes from the NSL KDD dataset are chosen. When evaluating the efficacy of the novel ensemble technique, classification accuracy, time spent developing the model, and the total number of false positives are considered. Experiments show that the Bagging ensemble achieves the maximum level of classification accuracy when working with a REP Tree base class. Creating the model in a shorter amount of time is one of the possible benefits that may be gained by using the bagging approach. The false positive rate generated by the ensemble approach is meager compared to that produced by other

machine learning methods.

Research on a Deep Neural Network Intrusion Detection System was carried out by Kim. et al. [30], using the KDD Cup 99 dataset produced by the whole AI community for use in the competition (DNN IDS). This was done due to the ever-changing nature of how network attacks are carried out. The data must first be prepared by undergoing transformation and normalizations before it can be input into the DNN model. In the preprocessing step, the data is first cleaned and then processed. Then, in the "learning model production" stage, the DNN technique is used on the data to create the model. In the last step, the model's accuracy is tested using the whole dataset from the KDD Cup 99. In the end, the accuracy, detection rate, and false alarm rate were calculated to determine the detection effectiveness of the DNN model, which has been demonstrated to deliver positive results in intrusion detection. This was done so that the detection effectiveness could be determined.

In the context of IDS, the proposed classification model addresses specific shortcomings in existing methods. Models aim to enhance detection accuracy, adapt to evolving threats, reduce computational complexity, improve scalability, enhance generalization, or provide interpretable results. The presentation of the new model is motivated by the need to contribute solutions to these challenges, ultimately advancing the effectiveness and efficiency of IDS in identifying and mitigating network intrusions. Challenges encompass issues such as false positives and negatives, the dynamic nature of the evolving threat landscape, handling massive volumes of real-time data, addressing imbalanced datasets, adapting to evolving network configurations, anomaly detection for novel attacks, resource constraints in IoT environments, privacy concerns in handling sensitive data, the complexity of multi-stage attacks, and scalability demands. These multifaceted challenges underscore the ongoing necessity for research and innovation in IDS to develop adaptive and robust solutions capable of effectively countering contemporary cybersecurity threats while addressing the complexities of diverse network environments.

3. Methodology

The methodology employed in this research involves developing and evaluating a particle swarm optimization (PSO) and AdaBoost-based intrusion detection system (IDS). The system encompasses several vital steps, including feature selection, training and testing of classification models, and performance evaluation using the NSL KDD dataset. The input dataset is subjected to the PSO algorithm for feature selection. PSO, a population-based optimization technique, mimics the behavior of a swarm of particles searching for the optimal solution. In this case, the algorithm is employed to identify the most informative features contributing significantly to accurate intrusion detection and classification. The particles converge toward the optimal set of features. After the feature selection process, multiple classification models are trained and tested on the dataset. A framework Fig. 1 typically illustrates the proposed model's flow of information and processes.

The models employed in this research include AdaBoost, K-Nearest Neighbors (KNN), and Naïve Bayes algorithms. These models are known for their effectiveness in detecting and classifying malware-related records. The AdaBoost algorithm, particularly, is utilized to build an ensemble of weak classifiers. Each weak classifier focuses on different aspects of the data and assigns weights to misclassified samples, improving overall classification accuracy. KNN and Naïve Bayes algorithms are also employed for comparative analysis to evaluate their performance against AdaBoost. The NSL KDD dataset, a well-established benchmark dataset for intrusion detection, is utilized for the experimental work. This dataset provides a realistic and representative sample of network traffic data, enabling a comprehensive evaluation of the proposed system. The performance of the PSO-AdaBoost IDS is evaluated based on various performance metrics, including accuracy, precision, and recall. These metrics provide insights into the system's ability to detect and accurately intrusion detect and classify intrusions

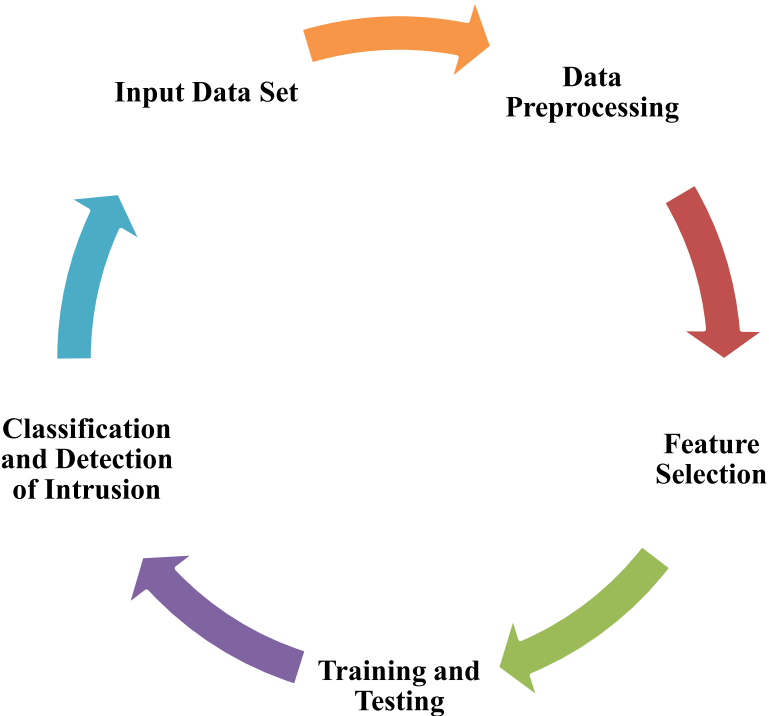


Fig. 1. PSO feature selection enabled optimized adaboost technique for efficient intrusion detection system for IoMT.

accurately. As flow mentioned in Fig. 2, the research findings indicate that the PSO-AdaBoost approach achieves the highest accuracy, recall accuracy, precision, and recall levels, validating its effectiveness in intrusion detection and classification tasks.

3.1. System implementations

In recent years, the rapid proliferation of interconnected medical devices and data-driven healthcare systems has led to a paradigm shift in contemporary healthcare, ushering in the era of the Internet of Medical Things (IoMT). However, this interconnected landscape has brought forth unprecedented challenges about the security and integrity of sensitive patient data and critical healthcare infrastructure. In response to these challenges, our designed system endeavours have been directed towards the meticulous development of an innovative and optimized

Intrusion Detection System (IDS), fortified with advanced machine learning capabilities, tailored explicitly for the intricate and dynamic environment of the IoMT. Our Designed system serves as an in-depth exploration of our research findings, detailing the intricacies of the design, components, and operational dynamics of our IDS, aimed at fortifying the security posture of the evolving landscape of contemporary healthcare technologies.

3.1.1. Innovative system architecture and design

In our research, we have comprehensively explored the intricate parameters governing the design of our Intrusion Detection System (IDS) within the dynamic realm of the Internet of Medical Things (IoMT). Leveraging insights from contemporary research, we meticulously engineered an innovative system architecture that seamlessly integrates a robust blend of advanced hardware components, including cutting-

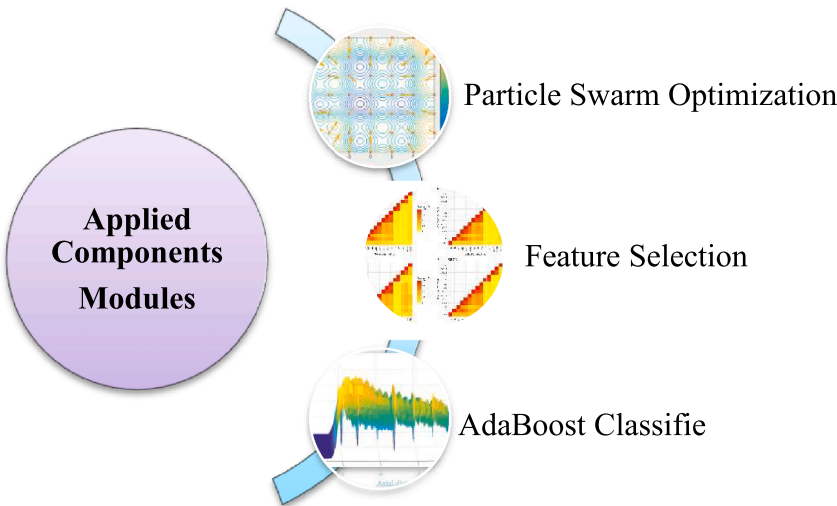


Fig. 2. Framework illustrates the flow of information and processes in the proposed model in the context of a combination of PSO and AdaBoost.

edge sensor networks and high-performance data processing units, with an intricately devised software framework as code mentioned in Supplementary File 1 (S1). This comprehensive design viewpoint ensures the seamless integration and interoperability of various medical devices, coupled with a resilient and scalable infrastructure tailored to safeguard the security and integrity of the IoMT ecosystem.

3.1.2. Integrated components and advanced modules

Building upon extensive insights gleaned from prior research endeavours, we meticulously incorporated integrated components and advanced modules within our IDS, essential for fostering a proactive and adaptive approach to intrusion detection and threat prevention. Our meticulously designed data preprocessing units, enriched with sophisticated data filtration techniques and anomaly detection algorithms, act as the initial gatekeepers, effectively screening and filtering incoming data streams. Additionally, our feature extraction modules are intricately calibrated to discern critical patterns and anomalies within the data, enabling effective threat detection and identification. Moreover, our IDS integrates a sophisticated ensemble of meticulously curated machine learning algorithms (Pseudocode as mentioned in S1), honed through extensive exploration, enabling real-time threat identification and proactive mitigation. These intricately woven components operate synergistically to fortify the system against potential security breaches and uphold the confidentiality and integrity of sensitive patient information within the complex web of the IoMT infrastructure.

3.1.3. Operational dynamics and adaptive learning capabilities

In alignment with the findings from extensive research studies, we have delved deep into the operational dynamics governing the machine learning model embedded within our IDS. Our comprehensive exploration sheds light on the model's adaptive learning capabilities, enabling it to adapt to the evolving patterns and swiftly discern potential intrusion instances based on sophisticated data patterns and anomaly indicators. Drawing from the wealth of insights from previous research studies, we highlight the model's adeptness in recognizing and categorizing diverse intrusion patterns and potential security threats with a high degree of accuracy and precision. Furthermore, our research emphasizes the model's agility and responsiveness to the dynamic shifts within the IoMT landscape. It underscores its capacity to swiftly adapt and respond to emerging cyber threats, establishing a robust and resilient security framework for contemporary healthcare technologies.

Through the meticulous integration of these advanced architectural

components, sophisticated modules, and comprehensive operational insights, we aspire to provide a complete and nuanced perspective on the intricate design and operational intricacies of our optimized IDS, thereby contributing significantly to the enhancement of cybersecurity practices and protocols within the evolving landscape of the Internet of Medical Things.

4. Results

The NSL KDD dataset [26], which contains 125,973 instances and 41 features, serves as the input dataset for this research. The dataset is divided into training and testing sets, with 20 % of the NSL KDD dataset allocated as training data, consisting of 25,192 records. The remaining 80 % of the dataset, amounting to 100,781 records, is used as testing data to evaluate the performance of the intrusion detection system. For the system's a feature selection process is conducted using the Particle Swarm Optimization (PSO) algorithm, resulting in the identification and selection of 12 relevant features for intrusion detection. In Fig. 3, we mentioned the detailed distribution of the NSL KDD dataset and presented the results obtained from the experiments. The figure likely showcases various aspects, such as the distribution of different attack categories and the frequency of each attack type. It potentially visualizes the performance metrics of the intrusion detection system for each attack category.

The results of the intrusion detection system using the NSL KDD dataset reveal its effectiveness in accurately identifying and classifying different attack types. The system detected attacks categorized as Denial of Service (DoS), aiming to restrict genuine users' services. Examples of DoS attacks include Smurf, Teardrop, SYN flooding, and Neptune. The system also demonstrated its capability in detecting User to Root (U2R) attacks, where the attacker gains control over local machines by exploiting vulnerabilities. U2R attacks include rootkit installation, spy activities, buffer overflow exploits, and SQL attacks.

Furthermore, the intrusion detection system effectively detected Root to Local (R2L) attacks involving unauthorised attempts to gain access to remote machines. Examples of R2L attacks encompass unauthorised access attempts, such as warezmaster activities, unauthorised IMAP access, multihoop traversal, and remote spying. Probe attacks, which involve gathering network information through traffic analysis, were also detected accurately by the system. Probe attacks include port scanning, satan network mapping, ping sweeps, and tools like Nmap. The successful detection and classification of these attack types

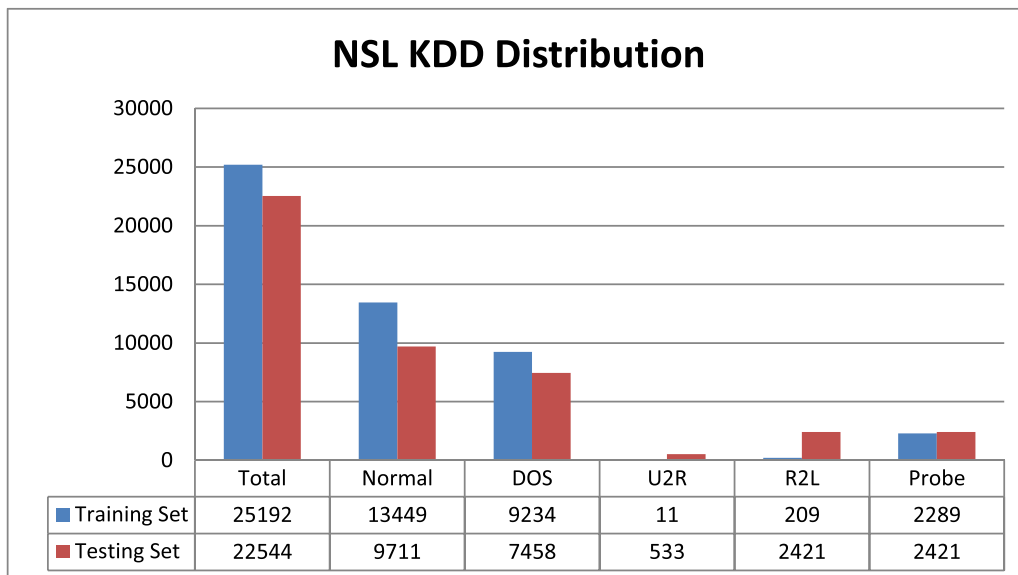


Fig. 3. NSL KDD data set distribution.

highlight the robustness of the intrusion detection system. By accurately identifying and categorizing different types of attacks, the system can provide timely responses and help mitigate potential security risks. The results obtained from the NSL KDD dataset demonstrate the system's ability to handle a wide range of intrusion attempts, ensuring the security and integrity of Internet of Medical Things (IoMT) devices and systems. The performance details of applied classifiers' accuracy, sensitivity, specificity, and precision of Classifiers for IoMT IDS are mentioned in Fig. 4.

The recall values of the applied classifiers for IDS (Intrusion Detection System) demonstrate their effectiveness in correctly identifying positive instances, specifically in the context of IoMT (Internet of Medical Things). Among the classifiers, Naive Bayes achieved a recall value of 0.883333, indicating that it successfully identified 88.33 % of the positive instances related to intrusions in the IoMT system. KNN (K-Nearest Neighbors) obtained a lower recall value of 0.282453, suggesting that it accurately identified only 28.24 % of the positive cases. On the other hand, AdaBoost showcased the highest recall value of 0.966667, indicating its exceptional performance in accurately detecting and classifying intrusions, capturing 96.67 % of the positive instances, as shown in Fig. 5. These results highlight the superior performance of AdaBoost in terms of recall, emphasizing its potential as a robust and reliable classifier for intrusion detection in IoMT systems. However, it is crucial to consider other metrics and factors alongside recall to comprehensively evaluate the performance of the classifiers, ensuring their suitability for real-world IoMT scenarios.

Evaluating an intrusion detection system's performance involves various metrics, as mentioned in Table 2. Accuracy, the most commonly used metric, measures the overall correctness of the system's predictions by considering both positives and negatives. Sensitivity, also known as recall, focuses on the system's ability to identify positive cases, aiming to

minimize false negatives. Specificity assesses the system's ability to identify negative issues, correctly minimizing false positives. Precision quantifies the proportion of accurate optimistic predictions among all positive predictions, reflecting the system's ability to minimize false positives. Recall, identical to sensitivity, measures the proportion of actual positive cases out of all positive patients, emphasizing the system's ability to identify true positives and avoid false negatives. Together, these metrics provide a comprehensive assessment of an intrusion detection system's performance, allowing researchers and practitioners to evaluate its accuracy, sensitivity, specificity, precision, and recall in accurately classifying and predicting different cases.

The accuracy of a classifier is a widely used metric to assess its success, representing the percentage of correct classifications it achieves. It is a key indicator when comparing different classifiers for a given task. While evaluating the fairness of classifiers is crucial, other factors are often overlooked. Multiple potential consequences may arise when conducting a test on a specific population or dataset associated with a particular condition. These outcomes include true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). To determine the utility of a classifier, it is crucial to strive for objectivity to the best of our ability. The metrics mentioned above, such as accuracy, sensitivity, specificity, precision, and recall, play a vital role in assisting this endeavor by providing comprehensive insights into the classifier's performance and ability to classify and predict different scenarios accurately. By considering these metrics, researchers and practitioners can make informed decisions and improve the effectiveness and reliability of the classifier in various applications.

5. Discussion

The Internet of Medical Things (IoMT) has emerged as a

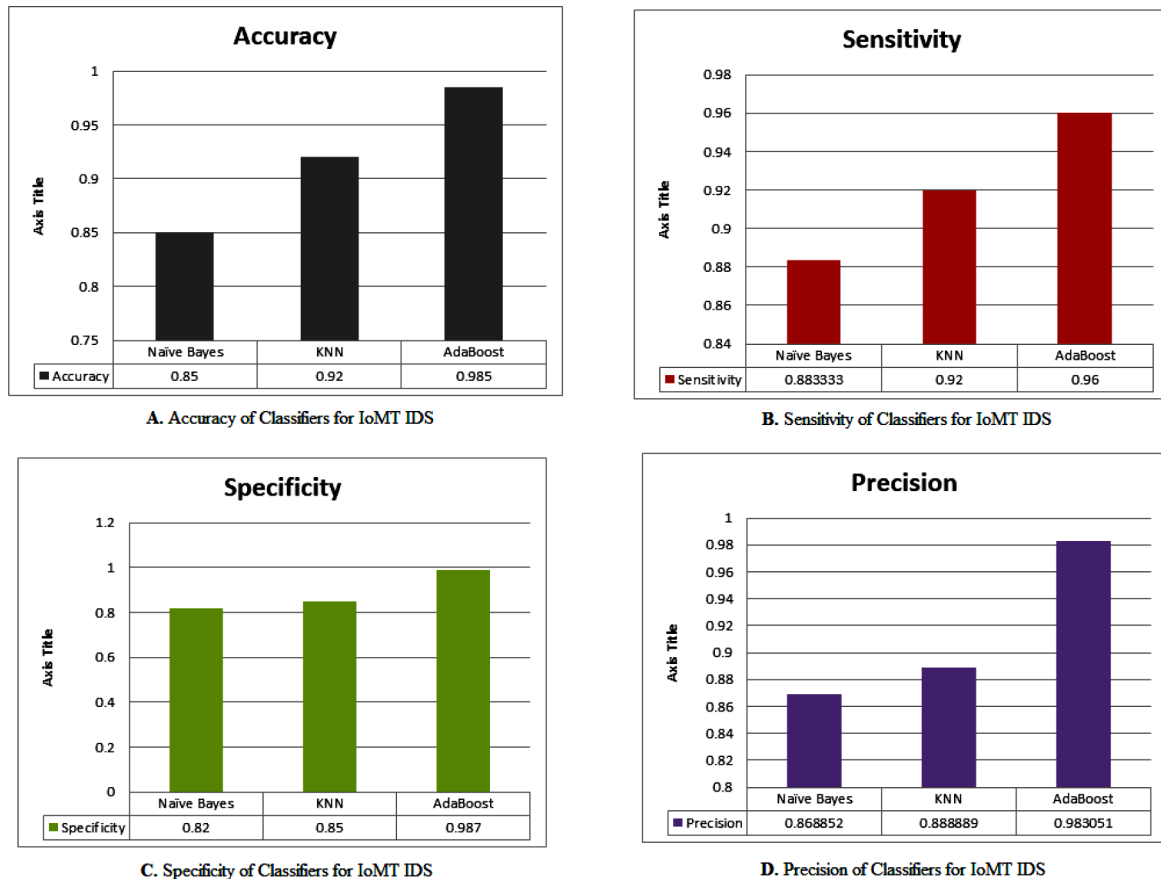


Fig. 4. Performance details of applied classifiers are (A) Accuracy (B) Sensitivity (C) Specificity and (D) Precision of classifiers for IoMT IDS.

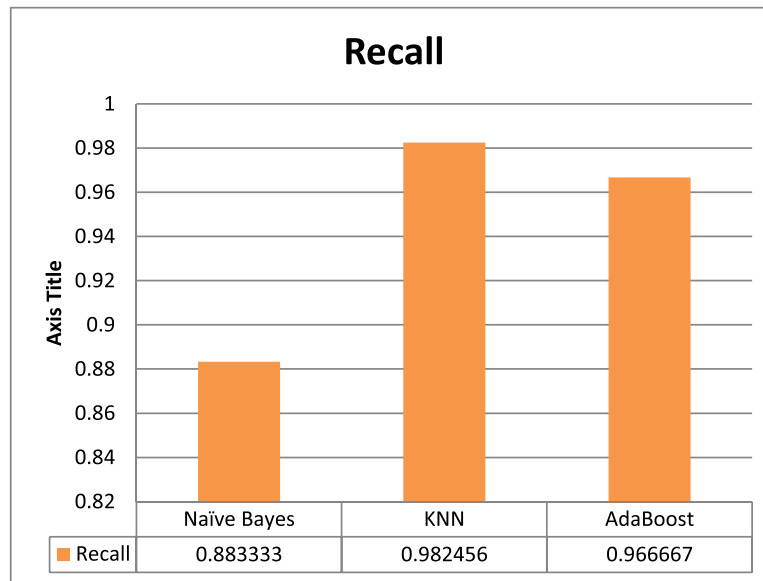


Fig. 5. Recall of Classifiers for IDS.

Table 2

Evaluation of an intrusion detection system's performance.

Metric	Calculation	Description
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Proportion of correct predictions overall predictions
Sensitivity	$TP / (TP + FN)$	The proportion of positive cases correctly identified
Specificity	$TN / (TN + FP)$	The proportion of negative cases correctly identified
Precision	$TP / (TP + FP)$	Proportion of true positive predictions among positives
Recall	$TP / (TP + FN)$	The proportion of actual positive cases correctly identified

transformative technology in the healthcare sector, offering the potential to enhance patient care, streamline operations, and improve healthcare outcomes [31]. However, the proliferation of IoMT devices and the exchange of sensitive medical data also introduce significant security challenges. Protecting medical data integrity, confidentiality, and availability is paramount to ensure patient privacy and maintain trust in IoMT systems [31]. In this context, our research focuses on developing an optimized machine learning-enabled intrusion detection system tailored to the unique requirements of IoMT. Traditional intrusion detection systems (IDS) often fall short in addressing the specific characteristics of IoMT, such as resource constraints, real-time data transmission, and the need for high availability [32]. Therefore, we propose an optimized approach that leverages machine learning techniques to effectively detect and mitigate intrusions in IoMT environments.

PSO [33] is an effective and efficient global optimization algorithm broadly utilized for pattern recognition and non-linear function optimization neural network training. In our results, we explore the process of searching for particles based on PSO, it consecutively adjusts its location towards global optimum by two factors: the best position identified by itself is specified by $P_i = (p_{i1}, p_{i2}, \dots, p_{iD})$ and that encountered by a complete swarm (gbest) is denoted as $P_g = (p_{g1}, p_{g2}, \dots, p_{gD})$ as mentioned in Fig. 6 and results were matched with earlier study by You. et al. [34]. Its velocity at the t th iteration is specified using $V(t) = (v_{t1}, v_{t2}, \dots, v_{tD})$. Equations evaluate the position at the next iteration. If an element of velocities exceeds the threshold or is set equal to the corresponding threshold. PSO computes from the scenario and utilizes it to solve optimization crises.

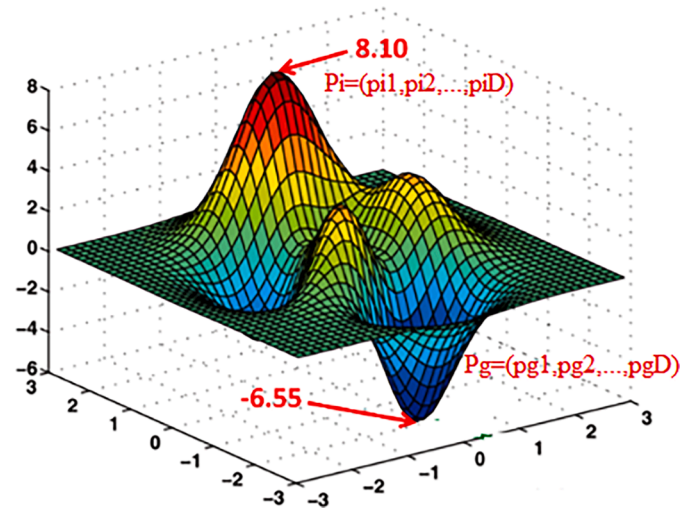


Fig. 6. PSO utilized for pattern recognition and non-linear function optimization neural network training.

In the context of Naïve Bayes [35], we explore and implement the Bayes' theory with an assumption of class conditional independence, which is highly reliable and frequently used in machine learning and data mining due to their ability to calculate probabilities for individual attribute-value combinations, making them suitable for various real-world issues amenable to computer-based diagnostics without the need for covariance matrices. Our results used the same classifiers: 'P' for probability, 'C' for class variables, and 'F' for feature variables, with the denominator being independent of 'C'. We matched our system's performance compared to earlier studies by Oğur et al. [37] and explore that Bayesian networks or DAGs (Directed Acyclic Graphs) employ graphical models to represent random variables and their conditional relationships with problems. We visualize probabilistic connections between variables, such as diseases and symptoms, and compute the likelihood of disease presence from a list of symptoms. We also comprise nodes representing hypotheses, including latent variables, observables, and unknown parameters, with edges connecting nodes that represent conditional dependencies as per the method of Tahsin et al. [36]. We also explore that in our system, each node incorporates a probability

function that calculates the likelihood of its parent variables given specific inputs.

In our results, we explore the implementation of the K-Nearest Neighbors (K-NN) classifier and check the organizational details of data for many investigations and problem interactions as per the method of Bermúdez et al. [37]. For pattern identification, we used a wide variety of pseudocode to assist in sorting and categorizing the input data. We calculate the distance based on K-NN classification between training samples, and our predicted events are shown in Fig. 7. We used locally estimated functions per the method of Liu et al. [38] and calculated the exact place on hold until after the classification process was finished. Results were compared with an earlier study [38]. Our results also categorize the least complicated input to apply when datasets are not known dispersed conditions and predict positive results. We explore that the pattern classification critical and K-nearest neighbor approach is valid for outstanding results when applied to various data sets.

Based on our results, setting K to 1 implies that the K-Nearest Neighbor (KNN) rule effectively functions as the Nearest Neighbor (NN) rule in its fundamental form and predict positive outputs. During our experimentation, we organized the samples into distinct clusters based on their shared features. This method facilitated a more structured approach in assessing and categorizing data points, allowing for informed estimations even when dealing with uncertain or ambiguous data instances. By utilizing a training set in conjunction with a query sample, we were able to establish a quantitative measure of the dissimilarity between disparate datasets. Consequently, this approach enabled us to accurately determine the identity of an unknown sample by contrasting it with the assigned cluster, providing valuable insights into the effective classification of data points within the scope of our research.

Based on our results and the insights gleaned from earlier research, AdaBoost emerges as a powerful gradient-boosting technique tailored for binary classification tasks, originating in the pioneering work conducted at the University of Michigan [39]. In our results, this method excels in constructing decision trees and continually refining their accuracy through iterative iterations compared to benchmark models. Particularly in the context of pedestrian detection sensors, AdaBoost's effectiveness shines through its image partitioning capabilities, enabling the systematic identification of pedestrians through a cascade of classification criteria, effectively enhancing the certainty of identification [40]. In our results, the Naive Bayes method is a fundamental approach for assigning class labels to problem occurrences based on feature value vectors [41,42]. To leverage the excellent results of our system, a diverse

array of techniques grounded in a joint base, Naive Bayes classifiers are trained through supervised learning, utilizing the maximum likelihood approach for parameter approximation [43] and predicting excellent results.

Analysis of the results derived from implementing the NSL KDD dataset highlights the efficacy of the intrusion detection system in accurately identifying various cyber intrusion types within the Internet of Medical Things (IoMT) environment. While the system demonstrates a comprehensive understanding of diverse attack categories, including Denial of Service (DoS), User to Root (U2R), Root to Local (R2L), and Probe attacks, certain limitations become apparent, such as the potential constraints in handling emerging and sophisticated cyber threats not adequately represented within the training dataset. The distinctive performance exhibited by different classifiers, particularly AdaBoost's superior recall performance, underscores the critical role of parameter selection in shaping the system's predictive capabilities. These findings emphasize the system's potential application in fortifying the security posture of interconnected medical devices, ensuring the integrity of sensitive patient data, and facilitating the seamless integration of advanced telemedicine and telehealth services within the evolving landscape of the healthcare sector.

Our optimization strategies play a crucial role in ensuring the practicality and efficiency of the proposed intrusion detection system. By fine-tuning parameters, reducing computational overhead, and optimizing the use of available resources, we have made it possible to implement our strategy on resource-constrained IoMT devices without compromising their primary functions as per the method of Nair et al. [44]. This optimization contributes to the feasibility of real-time intrusion detection in IoMT, a critical requirement in healthcare settings. Another important aspect of our research is the consideration of data privacy and compliance with healthcare regulations. We have addressed the challenge of securing medical data while in transit or at rest within the IoMT network, and the results were matched with earlier studies [31,45]. This protects patient privacy and ensures that healthcare organizations remain compliant with data protection regulations, such as the Health Insurance Portability and Accountability Act (HIPAA) in the United States.

Our research contributes significantly to the field of IoMT security by presenting an optimized machine learning-enabled intrusion detection system tailored to the unique challenges of this domain. We have demonstrated the potential for improved threat detection, resource efficiency, and compliance with data protection regulations. As IoMT continues to expand in healthcare, the findings and methodologies

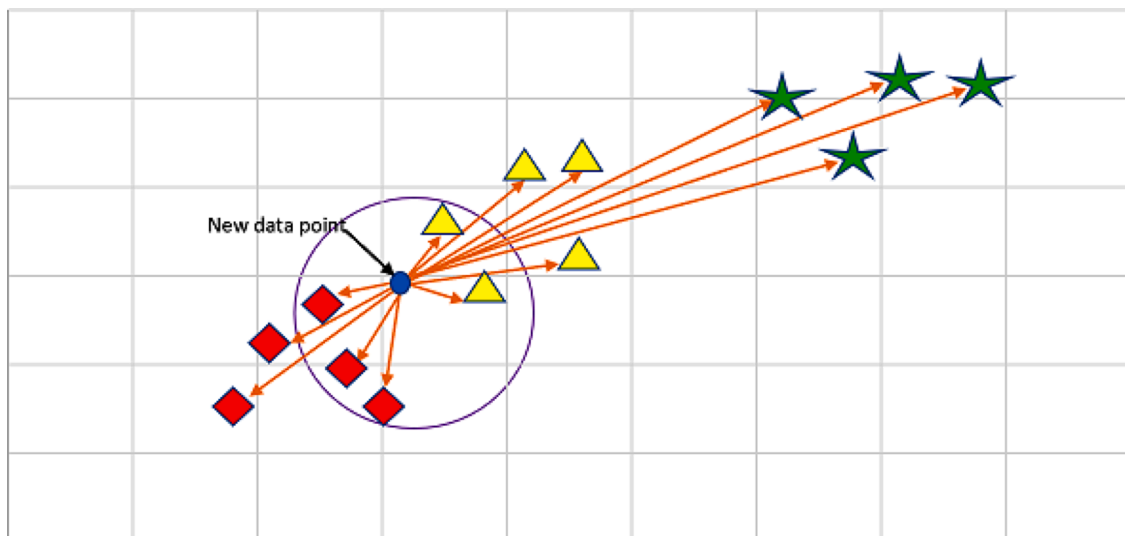


Fig. 7. K-NN classifier for a number of investigations.

presented in this research are promising to enhance the overall security posture of IoMT systems and ensure the continued trust of patients and healthcare providers in these technologies. Future research directions in this area may include further optimization techniques, the exploration of federated learning for distributed IoMT environments, and the development of automated response mechanisms to detect intrusions, thereby advancing the state of IoMT security even further.

6. Conclusion

We explore the remarkable potential of the Internet of Medical Things in enhancing healthcare for individuals requiring continuous medical care or monitoring. Integrating innovative health app platforms and incorporating machine learning-based intrusion detection systems are pivotal in delivering timely and efficient treatment while upholding patient privacy and securing sensitive medical data. ML-IDSs have gained prominence in computer security primarily due to their autonomous threat detection capabilities. In this research, we introduced an innovative intrusion detection system that harnesses the synergy of particle swarm optimization and AdaBoost algorithms. We utilized PSO for feature selection, and our classification models underwent rigorous training and testing using the NSL KDD dataset.

Additionally, we leveraged supplementary algorithms, including AdaBoost, KNN, and Naive Bayes, to effectively categorize intrusions. Our experiments yielded impressive results, with the PSO-AdaBoost approach demonstrating outstanding accuracy, recall accuracy, precision, and recall levels in detecting and categorizing intrusions. This underscores the system's efficacy in identifying potential security threats and proactively mitigating risks. Integrating ML-IDSs into innovative health app platforms can elevate patient care, reduce healthcare costs, and streamline the monitoring vital health indicators. Moreover, our unwavering commitment to patient privacy rights ensures the safeguarding of sensitive medical information, fostering trust and confidence among individuals who rely on continuous medical support. As we move forward, we must continue exploring innovative approaches and datasets to enhance the performance and reliability of ML-IDSs within the healthcare sector. By harnessing the power of machine learning and optimizing intrusion detection algorithms, we can take significant strides in fortifying the security infrastructure surrounding medical IoT devices, ultimately ensuring the well-being of patients in our increasingly interconnected world.

6.1. Implications of research

In the Internet of Medical Things (IoMT) context, this research carries profound implications by offering a blueprint for bolstering data security, preserving patient privacy, and ensuring compliance with healthcare regulations. As IoMT devices generate, transmit, and analyze extensive medical data, the research's recommendations are a vital framework for safeguarding this information, enhancing user trust, facilitating wider IoMT adoption, and nurturing patient-centric healthcare. By adhering to the research's security measures, IoMT platforms can fortify their resilience against cyber threats, streamline operations, promote innovation, and contribute to the global transformation of healthcare delivery.

6.2. Potential limitations and future research directions

The intrusion detection research limitations include data imbalance, generalization, scalability, and false positives. In contrast, future research may involve adaptive models, privacy-preserving techniques, federated learning, explainability, real-time detection, and device-specific models tailored to different medical IoT devices. These directions could help guide further research in the field of optimized machine learning-enabled intrusion detection systems for the Internet of Medical Things.

Data availability

Data used to support this study are available from the corresponding author upon request.

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CRediT authorship contribution statement

Zhenyang Sun: Writing – original draft, Investigation, Formal analysis, Writing – review & editing. **Gangyi An:** Writing – original draft, Investigation, Formal analysis, Writing – review & editing. **Yixuan Yang:** Writing – original draft, Investigation, Formal analysis, Writing – review & editing. **Yasong Liu:** Writing – review & editing, Supervision.

Declaration of Competing Interest

All authors declare that they have no conflicts of interest.

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All authors contributed to the study and agreed to be listed as authors.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.fraope.2023.100056](https://doi.org/10.1016/j.fraope.2023.100056).

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